

KIERAN SAUNDERS

+44 07305312004 ✧ UK

[Email](#) ✧ [LinkedIn](#) ✧ [Website](#) ✧ [GitHub](#) ✧ [Google Scholar](#)

PROFESSIONAL SUMMARY

PhD in Computer Vision with first-author publications at ICCV and BMVC. Expert in architecting and deploying robust deep learning systems for 3D reconstruction, monocular depth, and video understanding. Proven ability to lead research projects from concept to production-ready prototypes that achieve state-of-the-art performance on industry-standard benchmarks (KITTI, GraspNet). Seeking a Research Scientist role to solve complex 3D perception challenges in robotics, autonomous systems, or augmented reality.

PROFESSIONAL EXPERIENCE

Doctoral Researcher (Industrial Collaboration), Aurigo 2022 – Present

- Leading an industrial R&D project on a production-focused deep learning VSLAM system for autonomous airport baggage vehicles.
- Developed and benchmarked real-time 3D reconstruction systems to enable robust, on-device navigation for autonomous vehicles in GPS-denied zones.
- Created high-fidelity digital twins using neural rendering (NeRFs/Splatting) to enable realistic simulation for navigation testing and synthetic data generation.

Machine Learning Intern, Lampix (High-Achievers Programme) Winter 2021

- Selected from 200+ applicants for an intensive AI programme. Developed and deployed a deep learning model to precisely identify liquid levels from 2D camera feeds, powering an interactive projector-based augmented surface.

SELECTED RESEARCH & PROJECTS

RobustDepth: All-Weather 3D Perception ICCV 2023

- **Action:** Developed a robust depth estimation framework using a novel bi-directional pseudo-supervision loss, leveraging a custom-generated synthetic dataset to ensure high performance in adverse conditions.
- **Result:** Set a new state-of-the-art for all-weather depth estimation, reducing error by **39%** in fog (Foggy Cityscapes) and **30.5%** at night (NuScenes-Night) over previous methods, directly addressing a key challenge for autonomous vehicle safety.

BaseBoostDepth: State-of-the-Art Monocular Depth BMVC 2024

- **Action:** Engineered a novel curriculum learning strategy to leverage wider baselines in self-supervised monocular depth, overcoming common issues of occlusion and brightness inconsistency.
- **Result:** Achieved a new state-of-the-art on the KITTI benchmark (**AbsRel: 0.096**), outperforming previous methods. Significantly improved 3D reconstruction quality, with enhanced edge definition and point cloud fidelity on the SYNS-Patches dataset.

GRASP3R: RGB-Only 3D Grasping & Reconstruction Under Review

- **Action:** Architected the first end-to-end system to integrate dense 3D scene reconstruction with 7-DoF grasp prediction using only unposed, uncalibrated RGB images.
- **Result:** Demonstrated the viability of depth-free grasping by achieving competitive performance on the GraspNet-1Billion benchmark. The system's integrated 3D reconstruction enables effective collision checking, paving the way for lower-cost and more accessible robotic systems.

Dyna-DM: Efficient Dynamic Scene Understanding ICARSC 2023

- **Action:** Designed a lightweight, object-aware learning process that isolates truly dynamic objects from static ones in a scene to improve depth map accuracy and efficiency.

- **Result:** Improved depth estimation accuracy on the KITTI dataset while reducing GPU memory usage by 30%, demonstrating a clear path to more efficient on-device deployment.

TECHNICAL SKILLS

- **Core Expertise:** 3D Reconstruction, Monocular Depth Estimation, Visual SLAM, NeRF/Gaussian Splatting
- **Machine Learning:** Self-Supervised Learning (PyTorch), Generative Models (GANs, Diffusion), Real-Time Inference
- **Languages & Tools:** Python, C++, Git, Docker, SLURM, FFmpeg, Linux
- **Hardware & Sensors:** RGB/RGB-D Cameras, IMU, GPS; Multimodal Sensor Fusion

EDUCATION

PhD in Computer Science, Aston University 2022 – 2025 (Expected)

Thesis: Self-Supervised Monocular Depth Estimation.

Research part-sponsored by Aurigo, focusing on autonomous vehicle navigation and digital twin creation.

BSc in Mathematics, Aston University, First-Class Honours with Departmental Prize 2019 – 2022

PUBLICATIONS

- **Saunders, K.**, et al. "GRASP3R: Unposed and Uncalibrated RGB Robotic Object Grasping" (Under Review, 2025)
- **Saunders, K.**, et al. "BaseBoostDepth: Exploiting Larger Baselines For Self-supervised Monocular Depth Estimation" *BMVC*, 2024.
- **Saunders, K.**, et al. "Self-supervised monocular depth estimation: Let's talk about the weather." *ICCV*, 2023.
- **Saunders, K.**, et al. "Dyna-dm: Dynamic object-aware self-supervised monocular depth maps." *ICARSC*, 2023.